



Hydraulic fracturing; an Environmental issue- PSY-511

Environmental psychology (University of Haripur)

Hydraulic Fracturing; An Environmental Issue

Hydraulic fracturing (additionally **hydrofracturing**, **hydrofracking**, or **fracking**) is a well-stimulation approach wherein rock is fractured via a hydraulically pressurized liquid. High-pressure fluids (usually water-suspended chemicals and sand) are pumped into wells to create cracks in deep rock where natural gas, oil, and brine can flow more freely.

History

Hydraulic fracturing was first used in **Kansas** in 1947 in an attempt to extract natural gas from a limestone formation in the Hugoton gas field. Since that time, petroleum engineers have regularly used hydraulic fracturing as a means of increasing well production.

Uses

The technique of hydraulic fracturing is used to increase the rate at which fluids, such as petroleum, water, or natural gas can be recovered from subterranean natural reservoirs. Reservoirs are typically porous sandstones, limestones or dolomite rocks. Hydraulic fracturing enables the production of natural gas and oil from rock formations deep below the earth's surface.

While the main industrial use of hydraulic fracturing is in arousing production from oil and gas wells, hydraulic fracturing is also applied:

1. To stimulate groundwater wells.
2. To precondition or induce rock to cave in mining.
3. As a means of enhancing waste remediation processes, usually hydrocarbon waste or spill.

4. To dispose of waste by injection into deep rock formations.
5. As a method to measure the stress in the Earth.

Environmental Impact

The actual practice of fracking is only a small part of the overall process of drilling, completing, and producing an oil and gas well. However, since fracking involves injecting a chemical solution into the ground to free up oil and gas resources, there are some potential environmental impacts related to this process. These includes land use, risk of earthquakes, noise pollution, air emissions, water consumption, water contamination, risk of secondary accidents, and health effects on humans from environmental effects.

Significant amounts of land is used during hydraulic fracturing. About 3.6 hectares (8.9 acres) is needed per each drill pad for surface installations.

Noise pollution from hydraulic fracturing can affect residents and local wildlife; each well pad (in average 10 wells per pad) needs during preparatory and hydraulic fracturing process about 800 to 2,500 days of noisy activity, which affect both residents and local wildlife.

Air emissions are primarily methane that escapes from wells, along with industrial emissions from equipment used in the extraction process.

Hydraulic fracturing uses between **1.2 and 3.5 million** US gallons (4,500 and 13,200 m³) of **water** per well, with large projects using up to **5 million** US gallons (19,000 m³). Additional water is used when wells are refractured.

Water and air pollution are the biggest risks to human health from hydraulic fracturing; research is underway to determine if human health has been affected, and rigorous following of

safety procedures and regulation is required to avoid harm and to manage the risk of accidents that could cause harm.

Other Environmental Concerns

In addition to air and water pollution, fracking can have long-term effects on the soil and surrounding vegetation. The high salinity of wastewater spills can reduce the soil's ability to support plant life

Effects Of Hydraulic Fracturing On Health

Concern has been expressed over the possible long and short-term health effects of air and water contamination and radiation exposure by gas production. Health consequences of the crisis include infertility, birth defects and cancer.

A 2012 study concluded that risk prevention efforts should reduce air emission exposures for persons living and working near wells during well completions.

A survey conducted in Garfield County, Colorado and published in Endocrinology suggested that natural gas drilling operations may result in elevated endocrine-disrupting chemical activity in surface and groundwater.

Solutions

- Air pollution from fracking can be minimized by updating well equipment.
- Fracking's water consumption is its most dividing issue: fracking companies must reduce water use and properly treat their wastewater.

The ultimate solution to fracking is to abandon the practice, become less dependent on fossil fuels and increase investment in sustainable energy production. In the long-term, we must divert investments and subsidies from fossil-fuels into more sustainable energy sources.

05 Technologies And Methods That Can Make Fracking Clear

1. Using waterless fracking systems:

Traditional fracking systems use large amounts of water, therefore using water-free fracking systems can save a lot of water. This system may have other advantages, for example the company, GasFrac, uses gelled fluid containing propane instead of water in their fracking system. Using the fluid GasFrac uses can gather the same result as using water, but uses just one-eighth of the liquid and pumps the liquid at a lower rate.

2. Replace fresh water with recycled water or brine:

Besides using water-less fracking system, recycled water and brine also works perfectly in fracking system. Using recycled water and brine conserves freshwater as well as reduces the water pollution caused by traditional fracking systems.

3. Replace diesel powered equipment:

Diesel powered equipment produces poisonous pollutants and greenhouse gas when used to drill or pump wells. Therefore, replacing diesel-powered equipment with engines that use natural gas or solar panels will reduce the large amount of harmful gases emitted and lower the overall damage to the environment.

4. Introduce wastewater purification:

Hydraulic fracking produces more wastewater than the amount of oil it produces. The wastewater is usually shipped to special storage facilities that are underground, however,

treating wastewater can greatly reduce pollution caused by wastewater. The treated water can also be reused in the fracking system.

5. Reduce methane leaks:

One major concern of fracking is methane leaks. There are two improvements that can reduce methane leaks, one of them is using infrared camera that can detect leaks at fracking sites. The other improvement is replacing traditional pressure-monitoring pneumatic controllers to lower-bleed designs.